REACTIONS OF ORGANIC COMPOUNDS

PYQ • 2021 - 2026

1. 2026 **3 MARKS**

Some compounds are given.

$CH_3–COOH$, $CH_3–CH_2–OH$, $CH_3–CH_2–CH_2–COOH$,

$CH_3–CH_2–CH_2–COO–CH_2–CH_3$, $CH_3–CH_2–CO–CH_2–CH_3$

(a) Choose an ester from this.

(b) Write the IUPAC name of the acid and alcohol that must be reacted to prepare this ester.

(c) Write the equation for the chemical reaction of formation of this ester.

2. 2026 **2 MARKS**

(A) Some chemical reactions are given.

(i) $CH{\equiv}CH + HCl \rightarrow P$

(ii) $nP \rightarrow Q$

Write the structures and names of P and Q.

OR

(B) A chemical reaction is given. $CO + 2H_2 \rightarrow A$

(a) Write the structural formula of A.

(b) Write the chemical equation for the reaction that produces ethanoic acid from A.

3. 2025 **3 MARKS**

Chemical equations of some reactions are given.

(a) Which types of reactions are these ?

(b) Identify the product in the reaction (ii) and write any one use of it.

(i) $CH_3-CH=CH-CH_3+HBr \longrightarrow CH_3-CH_2-CHBr-CH_3$

(ii) $n\,CH_2=\underset{\displaystyle Cl}{\overset{\displaystyle |}{CH}} \longrightarrow \left[CH_2-\underset{\displaystyle Cl}{\overset{\displaystyle |}{CH}} \right]_n$

4. 2025 **2 MARKS**

Identify A and B.

(a) $CH_3–OH + CO \xrightarrow{\text{(catalyst)}} A$

(b) $CHCl_3 + Cl_2 \rightarrow B + HCl$

5. 2025 **1 MARK**

Which polymer is formed from isoprene ?

6. 2024 **3 MARKS**

Complete the following equations.

(a) ___X___ + $Cl_2 \rightarrow CH_3–CH_2Cl + HCl$

(b) $CH_3–CH=CH_2 + Cl_2 \rightarrow$ ___Y___

(c) $CH_3–CH_2–CH_3 \xrightarrow{\text{(Heat)}} CH_4$ + ___Z___

7. 2024 **2 MARKS**

Choose the right compounds from the given box to answer the following questions.

(i) $CH_3–CH_2–OH$ (ii) $CH_3–CH_2–COOH$ (iii) $CH_3–CH_2–CO–CH_3$

(iv) $CH_3–COOH$ (v) $CH_3–OH$ (vi) $CH_3–CH_2–COO–CH_3$

(a) Which among these is an ester ?

(b) Which are the compounds required to prepare this ester ?

8. 2024 **1 MARK**

Write the name of the byproduct obtained in the industrial production of soap.

9. 2022 **4 MARKS**

Choose the compounds from the box and answer the following questions.

CH_3COOH, $CH_3–COO–CH_2–CH_3$, $CH_3–CH_2–OH$, $C_{12}H_{22}O_{11}$

(a) Which is a carboxylic acid ?

(b) Which compound is an ester ?

(c) Identify ethanol.

(d) Which substance is used in the industrial preparation of ethanol ?

10. 2022 **3 MARKS**

(X) $CH_4 + 2O_2 \rightarrow$ _____ + $2H_2O$ + Heat

(Y) $nCF_2 = CF_2 \rightarrow [CF_2 – CF_2]_n$ (Teflon)

(a) Complete the chemical equation X.

(b) Name the reaction Y.

(c) Write any one use of Teflon.

11. 2021 **4 MARKS**

Find out the appropriate reactions and match the Columns A, B and C suitably.

Reactants (A) : $CH_3–CH_3 + Cl_2$; $C_2H_6 + O_2$; $nCH_2=CH_2$; $CH_3–CH_2–CH_3$

Products (B) : $CO_2 + H_2O$; $[CH_2–CH_2]_n$; $CH_2=CH_2 + CH_4$; $CH_3–CH_2Cl + HCl$

Name of Reaction (C) : Thermal cracking ; Substitution reaction ; Combustion ; Polymerisation

12. 2021 **2 MARKS**

Given below are two chemical equations.

(a) $CH_2=CH_2 + H_2 \rightarrow A$

(b) $A + Cl_2 \rightarrow$(sunlight) $B + HCl$

Identify the compounds A and B.

13. 2021 **1 MARK**

Select the monomer of natural rubber.

[Vinyl Chloride, Ethene, Isoprene, Tetrafluoro ethene]